

BabyShopHub

Test Plan

Aptech eProject Submission

Version 1.0

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1. Purpose and scope

This Test Plan defines how BabyShopHub is verified against its requirements. It sets out the testing strategy, the environment, a table of concrete test cases covering every functional area, the entry and exit criteria for a verified release, and how results are recorded.

In scope: every functional requirement FR-1 to FR-27 from the SRS, and the security and error-handling non-functional requirements that can be exercised through the UI.

Out of scope: real payment processing (checkout is simulated), any network or multi-device behaviour (the app is entirely local), and formal accessibility certification (which requires assistive-technology testing beyond this project).

2. Testing strategy

Testing proceeds in two tiers, from cheapest to most authoritative.

Tier 1 — Static analysis. The project is checked with flutter analyze, the Dart analyzer, which type-checks every file and flags invalid references, dead code, and lint violations. The rules are configured in analysis_options.yaml (the standard flutter_lints set). It should report No issues found!. This is the first gate and is run after every change.

Tier 2 — Manual functional testing. With the app running (flutter run -d windows), the test cases in Section 4 are executed by hand against the seeded demo data, following the flows in the User Guide. Each case lists the steps to take and the result to expect.

This two-tier approach is deliberate: static analysis catches code-level errors early and cheaply, and manual testing confirms that the running app behaves the way the requirements specify.

3. Test environment

Item	Value
Platform	Windows desktop (flutter run -d windows), or Android device/emulator
Build	Debug build via flutter run
Data	The database seeded on first launch (see Developer Guide §6)
Customer account	parent@babyshophub.com / Parent@123
Admin account	admin@babyshophub.com / Admin@123
Reset between runs	Delete the local database file (app rebuilds seed data)

4. Functional test cases

Each case lists what to do and what should happen, with the requirement it covers. IDs are grouped by area.

4.1 Authentication and accounts

ID	Test	Expected result	Req
TC-01	Register with valid details	Account created, signed in, Home shown	FR-1
TC-02	Register with a weak password (e.g. "abc")	Rejected with a validation message; no account created	FR-1, NFR-4
TC-03	Register with an email already in use	Rejected with a clear "email already registered" message	FR-1
TC-04	Log in with correct customer credentials	Storefront (customer shell) opens	FR-2
TC-05	Log in with correct admin credentials	Admin panel opens	FR-2
TC-06	Log in with a wrong password	Rejected with an error; not signed in	FR-2
TC-07	Close and reopen the app while signed in	Session restored; returns to the correct home	FR-3
TC-08	Sign out	Returns to login; session cleared	FR-4
TC-09	Edit profile name/phone; add and default an address	Changes saved and reflected at checkout	FR-5

4.2 Catalogue, search, and reviews

ID	Test	Expected result	Req
TC-10	Open Categories and enter one	Products in that category are listed	FR-6

TC-11	Search by product name	Matching products shown	FR-7
TC-12	Search by brand	Matching products shown	FR-7
TC-13	Apply price/rating/a availability filters and a sort	Result set narrows and reorders accordingly	FR-7
TC-14	Open a product detail	Image, price, brand, stock, seller, rating, reviews shown	FR-8, FR-9
TC-15	Write a review on a delivered product	Review appears; product average updates	FR-10
TC-16	Rate a seller	Seller rating recorded separately from product reviews	FR-10

4.3 Cart, checkout, and payment

ID	Test	Expected result	Req
TC-17	Add a product to the cart	Cart badge increments	FR-11
TC-18	Increase quantity beyond stock	Plus button stops at available stock	FR-11, NFR-4
TC-19	Change quantity and remove an item in the cart	Totals recompute; item removed	FR-12
TC-20	Observe free-delivery hint	Cart shows how much more to add; delivery becomes free above threshold	FR-12
TC-21	Proceed to checkout	Order summary shows items, address, and price breakdown	FR-13
TC-22	Enter a malformed card number	Rejected with a validation message	FR-14, NFR-4
TC-23	Complete checkout with a well-formed card	Order confirmation with reference; cart emptied	FR-14, FR-15

4.4 Orders and tracking

ID	Test	Expected result	Req
TC-24	Open Orders after placing one	New order listed with status Pending	FR-16
TC-25	Open an order's detail	Tracking timeline with dated status stages shown	FR-17
TC-26	Cancel an order before Shipped	Order becomes Cancelled; Cancel hidden once Shipped	FR-17

4.5 Support

ID	Test	Expected result	Req
TC-27	Read the FAQ	Help topics displayed	FR-20
TC-28	Open a support ticket	Ticket created in Open status	FR-18
TC-29	Exchange messages on a ticket	Messages appear in the conversation	FR-19

4.6 Administration

ID	Test	Expected result	Req
TC-30	Open the admin dashboard	Summary counts/figures shown	FR-21
TC-31	Create a product with valid data	Product added; visible in storefront	FR-22
TC-32	Save a product with an empty name or bad price	Rejected with a validation message	FR-22, NFR-4
TC-33	Advance an order Pending → Confirmed → ... → Delivered	Only valid next steps offered; each recorded on the timeline	FR-24
TC-34	Delete a category that still has products	Refused with an explanation	FR-23
TC-35	Delete an	Removed	FR-23

	empty category		
TC-36	Suspend a customer, then attempt login as them	Login refused while suspended; restore re-enables	FR-25, NFR-5
TC-37	Attempt to suspend the last active admin	Refused to protect management access	FR-25
TC-38	Hide, restore, and delete a review	Hidden review leaves the storefront and rating; delete removes it	FR-26
TC-39	Reply to a ticket and change its status	Reply reaches the customer; status updates (incl. reopen)	FR-27

5. Non-functional test cases

ID	Test	Expected result	Req
TC-40	Time common actions (open screen, search, add to cart)	Each completes within ~ 1–2 seconds; no network wait	NFR-1
TC-41	Trigger validation errors across forms	Every failure shows a clear, human-readable message; no crash	NFR-4
TC-42	Inspect stored password (e.g. via the DB file)	Stored as a salted SHA-256 hash, never plain text	NFR-5
TC-43	Attempt to reach admin features as a customer	Not reachable; admin shell is never presented to a customer	NFR-5
TC-44	Run flutter analyze	Reports No issues found!	NFR-7

6. Entry and exit criteria

Entry criteria. The app builds and launches (Setup Guide completed), and the seeded demo data is present.

Exit criteria. The release is considered verified when:

- all functional cases TC-01 to TC-39 pass;
- the core journeys pass end to end — customer register-to-order (TC-01, TC-17, TC-21, TC-23) and admin order fulfilment (TC-33);
- the security cases pass — salted-hash storage (TC-42), access separation (TC-43), suspended-login refusal (TC-36), last-admin protection (TC-37);
- flutter analyze reports no issues (TC-44).

Any failing case is logged with its ID, the steps, the observed result, and a screenshot, and is re-tested after a fix.

7. Recording and reporting results

Each test case is run in order and its outcome recorded against its ID as **Pass** or **Fail**. A convenient way to keep the record is a simple table:

Field	What to note
Test ID	The case run, e.g. TC-23
Result	Pass or Fail
Notes	For a failure: the exact steps, what was expected, and what happened
Evidence	A screenshot of the screen at the point of interest

A failing case is logged with these details, fixed, and then re-run to confirm the fix before the release is considered verified against Section 6.

Executing TC-01 through TC-44 against the running app completes verification of the requirements defined in the SRS.